Aryan Goel

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EDUCATION

University of Southern California — Los Angeles, CA
Master of Science, Computer Science (Data Science Specialization) — *Aug. 2025 – May 2027 (Expected)*
***Relevant Coursework**: Machine Learning for Data Science, Applied Statistical Inference, Deep Learning, Algorithms, Database Systems (indexing, storage engines, query optimization), Foundations of Data Management (Hadoop, Spark, distributed SQL), NLP*

Manipal University Jaipur — Jaipur, India
Bachelor of Technology, Computer & Communication Engineering — GPA: 9.15/10 — *July 2021 – May 2025*

TECHNICAL SKILLS

Languages: Python, SQL, R (basic)
Statistics & Experimentation: Hypothesis Testing, ANOVA, A/B Testing, Power Analysis, Confounding/Simpson's Paradox Analysis, Causal Reasoning
ML & Modeling: Scikit-learn, PyTorch (CNNs), Logistic Regression, Random Forest, Gradient Boosting, Class Imbalance (SMOTE), Feature Engineering
Explainability & Evaluation: SHAP, Grad-CAM, Precision/Recall/ROC-AUC, Sensitivity-Weighted Metrics for Imbalanced Data
NLP & GenAI: Hugging Face Transformers, Text Summarization (ROUGE), Tokenization & Embeddings, Multi-Agent Orchestration (AutoGen)
Data & Databases: Pandas, NumPy, SQL (joins, window functions, query optimization), PostgreSQL, DuckDB, Indexing & Storage Fundamentals
Tools & Cloud: Matplotlib, Seaborn, Git, AWS EC2, GCP (Vertex AI), Streamlit

EXPERIENCE

Indian Institute of Remote Sensing (ISRO) — Dehradun, India
Research Intern — *Feb. 2025 – May 2025*

- Engineered a cascaded CNN architecture (CED-Net) for crop/weed segmentation on RGB drone imagery, improving accuracy by **[X%]** over baseline U-Net via residual blocks and attention mechanisms
- Built the full annotation-to-evaluation pipeline (LabelMe → pixel masks → training/eval on **IoU and Dice score**), benchmarking multiple architecture variants

Indian Institute of Technology (IIT) Roorkee — Roorkee, India
Research Summer Engineer — *May 2024 – July 2024*

- Devised a **Physics-Informed Gaussian Process Regression** model to forecast flood events, improving forecast reliability by **[X%]** over a standard GPR baseline by embedding hydrological constraints
- Executed **uncertainty quantification** via Monte Carlo and Multi-Level Monte Carlo simulations for confidence-bounded risk predictions

Indian Institute of Technology (IIT) Roorkee — Roorkee, India
Software Developer — *June 2023 – July 2023*

- Managed and optimized geospatial data layers in **GeoServer** for a production WebGIS application; performed data classification with QGIS
- Redefined the rendering pipeline and resolved performance bugs, cutting load time by **[X%]** while shipping new features

KEY PROJECTS

Telecom Churn Prediction & A/B Test Analysis | *Python, Pandas, Scikit-learn, SciPy*

- Uncovered a **Simpson's paradox**: a support feature appeared to increase churn in aggregate (**+8pp**) but genuinely reduced churn by **30-40% relative** within every contract-type segment — traced to self-selection bias, not causation
- Cleaned inconsistent labels, mixed-type currency fields, and missing data across multiple columns; compared Logistic Regression, Random Forest, and Gradient Boosting on precision/recall/ROC-AUC
- Designed and simulated an **A/B test** for a retention-offer intervention, including a priori **power analysis** and detection of an embedded novelty effect

Tumor Detection with Explainable AI (XAI) | *PyTorch, SHAP/Grad-CAM, GCP (Vertex AI)*

- Built an end-to-end deep learning pipeline for tumor detection on medical imaging data, deploying the trained model for inference on **GCP**
- Integrated **Explainable AI** (Grad-CAM/SHAP) to visualize which image regions drove each prediction, addressing the interpretability bar required for clinical deep-learning applications
- Evaluated with recall/sensitivity-weighted metrics appropriate for imbalanced medical data (minimizing false negatives), achieving **[recall/AUC score]**

BrandSync: Agentic Multi-Modal Ad Generation | *Supabase, FastAPI, Google ADK, Gemini Pro (team project)*

- Built the caching and feedback-routing layer for a multi-agent creative pipeline, reducing redundant regeneration and cutting downstream API calls by up to **80%** on targeted revisions — the routing logic was framed as a cost-minimization problem over which pipeline stages actually needed to re-run

ADDITIONAL PROJECTS

- **2-Stage Loan Approval & Valuation System —** imbalanced-data classifier using SMOTE, with evaluation framed around false-negative cost (bad loans approved) rather than raw accuracy *(Python, Scikit-learn)*
- **Text Summarization with Hugging Face Transformers —** abstractive summarization pipeline fine-tuned/applied on transformer models, evaluated with ROUGE *(Hugging Face, PyTorch)*
- **AutoGen Data Analyzer GPT —** multi-agent data-analysis system where agents collaboratively write, execute, and critique analysis code against a live dataset *(AutoGen, Python, LLM API)*

ACHIEVEMENTS

- 1st Place, Inter-College Hackathon (Mental Health track) — built a Figma-prototyped app with an integrated chatbot
- Dean's List for Academic Excellence, Manipal University Jaipur (awarded twice)